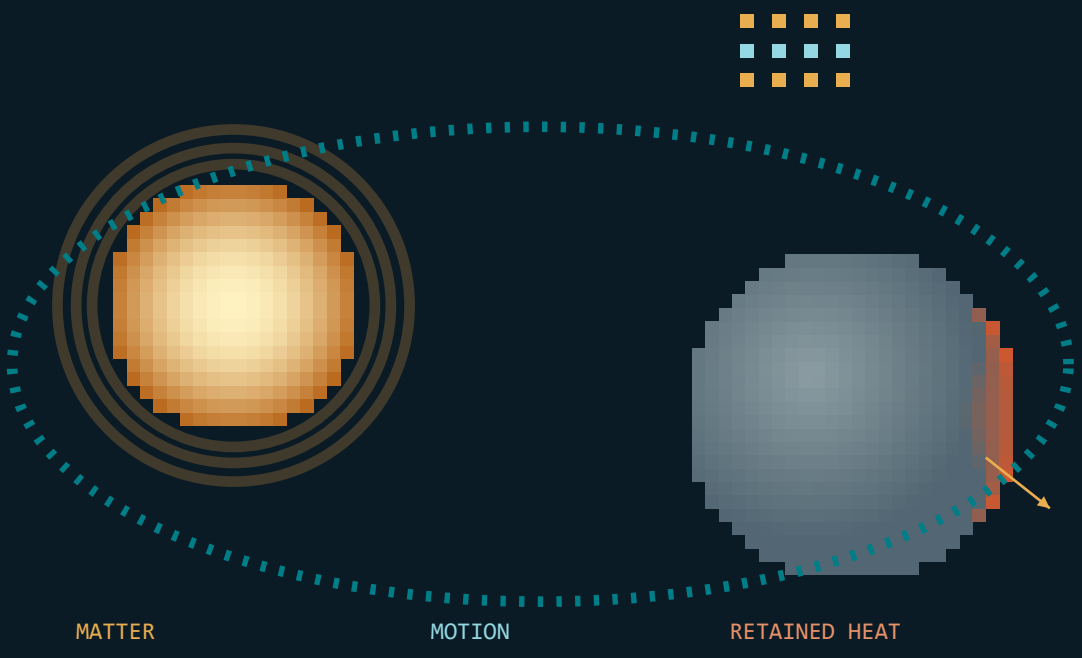
PROJECT KUJIRA

The completed v19 engine

PHYSICAL INVENTORY / COARSE TIME / EXACT PARALLEL WORK



An illustrated reference to the models, ownership rules, rendering and release evidence behind milestones M1-M4.

INSTALLED 19.9.1 / SAVE FORMAT 37

GitHub @kujiwa-rgb / X @officialemane

29 October 2026 / Original project material copyright 2026

A new reference, not a rewritten archive

Read current v19 behavior without mistaking an older proposal for a delivered model.

Collection item	Role in this edition
Universal Topology IV / v16.3.1 / 52 pages	Historical atlas. Its earlier units, materials and visual language provide context, not the current save or allocation contract.
v16.6 Model Update Addendum / 18 pages	Historical changes to light, wind, stellar appearance and the magnetic proxy.
Planetary Surface and Subscale Physics / 12 pages	Design proposal, not an installed patch. Its 325-km artwork and proposed local tile are reconciled here.
Universal Topology V / this volume	Current v19 reference, ending at installed 19.9.1 and save format 37.

Evidence hierarchy

Frozen receipts and final logs outrank stage notes. Later release records outrank earlier plans. A model diagram explains a convention; it is not a measured field solution.

Status language

Implemented means reported delivered behavior. **Approximation** means an explicit numerical/gameplay convention. **Measured** is tied to a named fixture and machine.

Scope

This is a v19 edition. It does not invent complete v17-v18 release histories to bridge the older atlas. Earlier volumes remain preserved.

Documentation only

No engine source, installed binary or player save is changed by preparing this volume. The recorded release evidence is not a fresh engine regression run.

> **Important supersession**
> A 375-km raster reference is not the current physical birth minimum. New solid bodies must be at least 1 km, and all independent solids use the shared body pool.

The v19 model map

Forty pages, from ownership to the installed 19.9.1 closeout.

Pages	Topic	What to look for
04-06	Milestones and ownership	M1-M4, retired proposals, capacities and the 1-km exception
07-12	Matter, motion and contact	Units, gravity, pair potential, ejecta and population storage
13-18	Thermal and atmosphere	Reservoirs, phase, light, clouds, drag and entry heat
19-22	Charge, fields and shadows	Paid charge, gas lightning, magnetic convention and finite sources
23-25	Stellar Time	Coarse future state, linked motion, contacts and actual completion
26-29	Surface and observation	History, subsurface colour, raster refinement and Tab steering
30-31	Exact performance work	Persistent pool, ordered sums and measured costs
32-34	Persistence and release	Format lineage, gate, limits and owner-approved exceptions
35-36	Sources and review	Release files, evidence, code map and bounded review questions
37-40	19.9 closeout	Vapour display, paid ejecta, measured impacts and installed identity

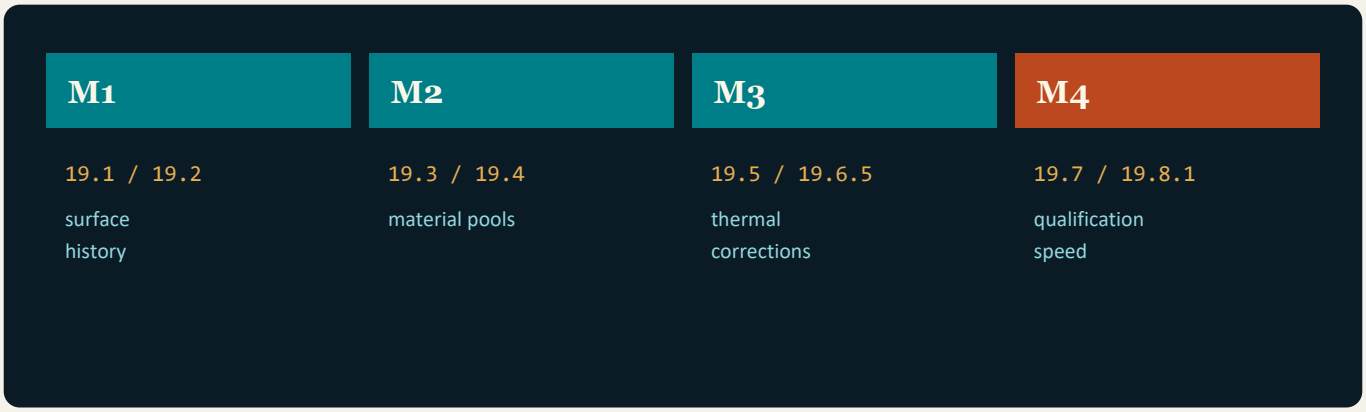
One central idea

Material keeps its physical inventory and history at every zoom. Grouping is not physical merging. Rendering reads the result of the solver, rather than creating a second universe.

Conservation has a declared boundary
Ordinary transfers conserve the supported inventory and energy ledgers. The isolated sub-1-km retirement policy deliberately removes material and heat; that narrow gameplay exception must stay visible.

From surface view to retained material

M1 and M2 established observation, history and bounded storage.



Milestone sequence, not four additional universes or independent material copies.

Part	Delivered progression	Save
19.1	Read-only surface view to zoom 3,750; universal 10-km navigation triangle; body-fixed exterior clearance.	26
19.2	64-column surface records, deposits/seams/damage, history transfer through mergers and fragments; watchdog.	27
19.3	Packets and four-mode bins; body/material gravity from detachment; coarse orbital paths; cosmetic sampling.	28
19.4	512 packet records; loose layer, landing classes and sliding; material encounters; staged loading and crash log.	29
19.4.1-2	Preset/view corrections, one refining raster, locked focus, triangle-anchored zoom and contact-based heat placement.	29-30

Read M2 as history where superseded

Early descriptions of small solid populations in packets/bins do not authorize that runtime today. 19.6.4 moved independently positioned solids into the shared 1,024-body pool.

M3 corrections, then M4 qualification

Completion includes corrections and measured boundaries, not just a version label.

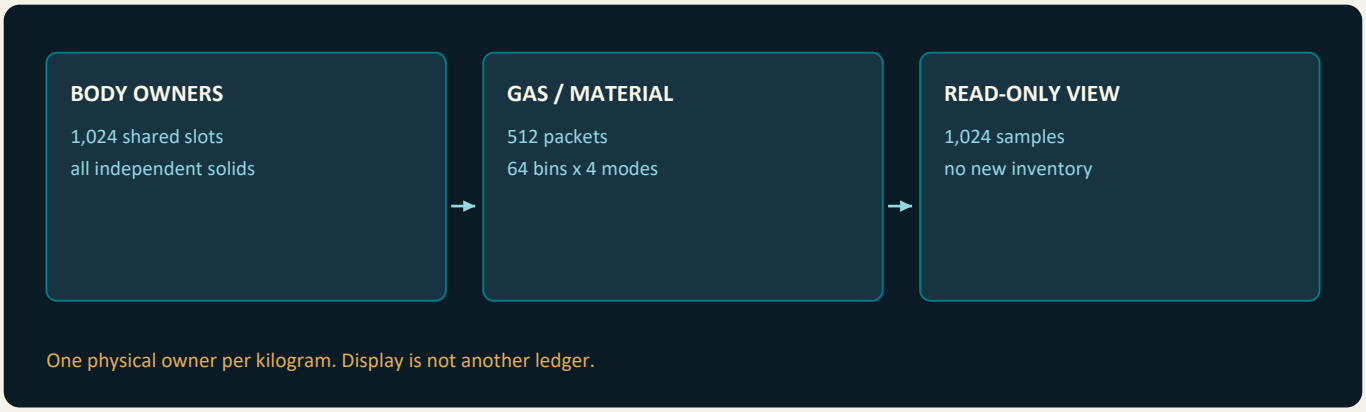
Part	What changed
19.5	Packet/bin thermal continuity; cloud ownership, resting columns, rain and drag. Starlight only heats. Owed contact half-kick. Glow from emitting skin. Explicit focus.
19.6.0-2	Paid charge and eligible gas lightning; Tab; thermal-colour protection; steering, shape, debris visibility and guarded launcher corrections.
19.6.3	Persistent IDs, contact/motion integration, finite-source light, linked REBOUND and a distinct coarse Stellar Time scheduler; cold-state stall correction.
19.6.4-5	Unified main-body solids; conservative migration/refusal; authoritative recoil remainder; local entry/impact heat; tapered subsurface colour.
19.8.0	M3 math review; stable two-sign charge root; conservative contact x-sweep; exact per-item parallel passes; 19.7 combined long-time case.
19.8.1	Persistent workers; shared exact pair-potential helper; parallel packet potentials and field-light passes. Molten-heavy tick accelerated without a new physical approximation.
19.9.0-1	Directional bounded debris; vapour packets and unified gas-layer display; unequal power-law pieces, impact-funded speeds and eligible paid chip escape.

Plans that did not become the engine

- The proposed 128 x 64 local fluid tile was replaced by bounded material and atmosphere representations.
- No new electromagnetic terrain, universal shield or general MHD solver was delivered by M3.
- Composite contact-binary bodies remain unsupported; supported solid mergers become one body.

One owner, several bounded representations

The numbers 1,024, 512 and 256 describe different allocations.



Allocation	Current rule
1,024 main-body slots	All stars, planets, spawned bodies and independent solid products share these slots. No additional solid ceiling of 256.
512 packets; 64 x 4 bin modes	Separate supported gas/material storage. Not a hidden population of disconnected solid rocks.
1,024 samples; 1,024 surface records	Samples are shared cosmetic output; surface records describe existing owners and history.
256 weather owners; 256 charge regions	Weather allocation and separated electrical state are not body counts.

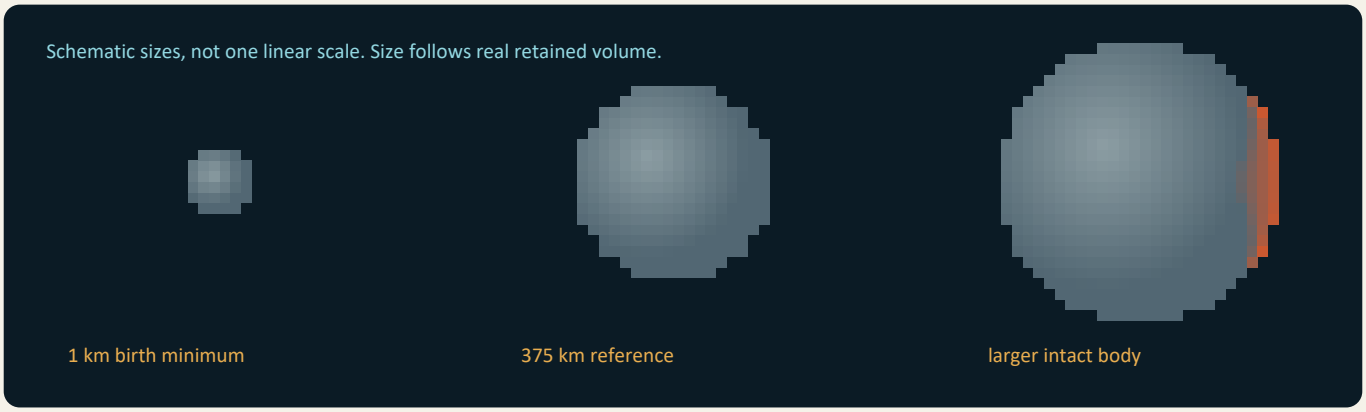
1-km birth rule and the cleanup exception

Fund every new solid from actual retained volume and count. Refuse unsafe, undersized or unaffordable births before donor mutation. For an existing isolated solid that falls below 1 km, owner-approved retirement removes its contents and energy. Actual contact uses the supported handler instead.

SOURCES S01, S06 / register pp.35,40

Size comes from retained volume

A pixel is not a kilogram, and a display reference is not a physical floor.



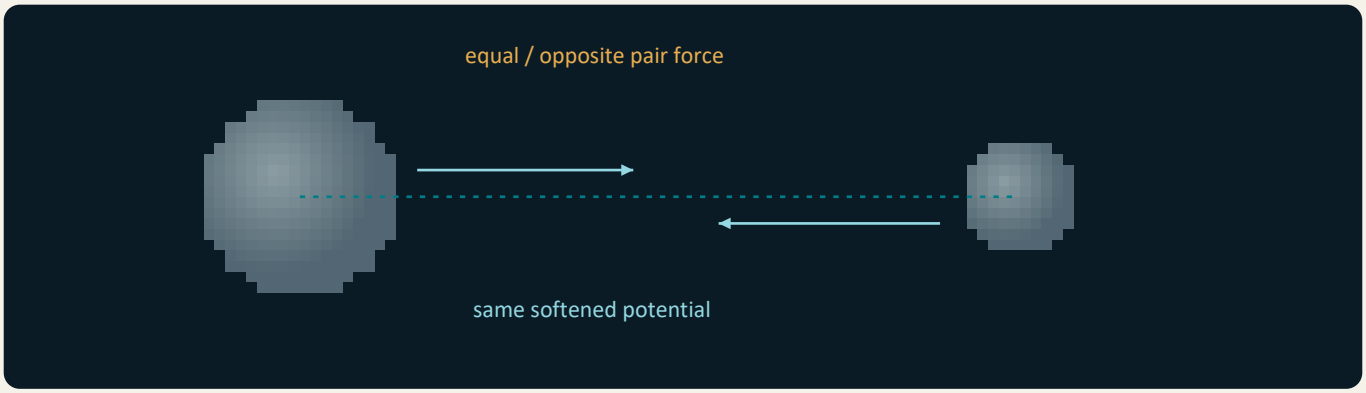
$$V = \Sigma_k \, m_k \, / \, \rho_k(T, p, \text{phase})$$ 7.1

$$d = 2 \, [3V \, / \, (4\pi)]^{1/3}$$ 7.2

- **World scale:** 3,000 km per world unit; 3,000,000 m per world unit. Ordinary live ticks represent 10 simulated seconds.
- **Material response:** supported constituent and layer densities depend on the existing temperature/pressure/phase model. A changed condensed radius does not itself prove lost retained mass.
- **Geometry:** N disconnected pieces are not one connected equivalent-volume sphere. Halving diameter at equal density divides mass by eight.
- **Reference scale:** 375 km is the older chip/raster reference. The surface-design PDF used 325 km as artwork; it is not a new engine constant.

Matched gravity, potential and derivative

The exterior softened point-source form is not every stellar interior.



$$a_r = -GM\, r\, (r^2 + \varepsilon^2)^{-3/2} \qquad 8.1$$

$$\Phi = -GM / \sqrt{(r^2 + \varepsilon^2)} \qquad 8.2$$

$$da_r/dr = GM\, (2r^2 - \varepsilon^2) / (r^2 + \varepsilon^2)^{5/2} \qquad 8.3$$

World acceleration uses $G_{world} = G_{SI}$ / meters_per_unit^3. Local-size softening applies to supported planetary pairs; distributed gas uses its existing extent rule. Stars and remnants retain their matched stellar kernels.

Recorded approximation
Bodies interact gravitationally with packets and bins. Packet-packet, packet-bin and bin-bin self-gravity are omitted. The engine is not a fully resolved all-material N-body continuum.

A changed body re-books its gravity

Changing mass, radius or a gravity profile changes more than temperature.

$$\Delta E_{bind} = \Sigma_o \left[U_{pair}(before,o) - U_{pair}(after,o)\right] \quad 9.1$$

The summation covers relevant other bodies and material records according to the existing helper and skip rules. Here U_{pair} means pair-potential **energy in joules**, not the specific potential Φ in J/kg on page 8.

Matched interiors

Red giants and supergiants use a structured core plus envelope profile. Force, potential and derivative are matched at supported joins; the exterior expression alone is not a substitute for this model.

19.8.1 exact helper

Six copied re-booking loops now call addPairPotentialChange. Independent pair terms can run in parallel, but each is added to binding in the same original body order and from the same starting value.

Settlement

When phase, contraction or a material transaction changes geometry, the pair ledger is settled against the changed owner. Rotation and thermal energy also retain their existing transactions.

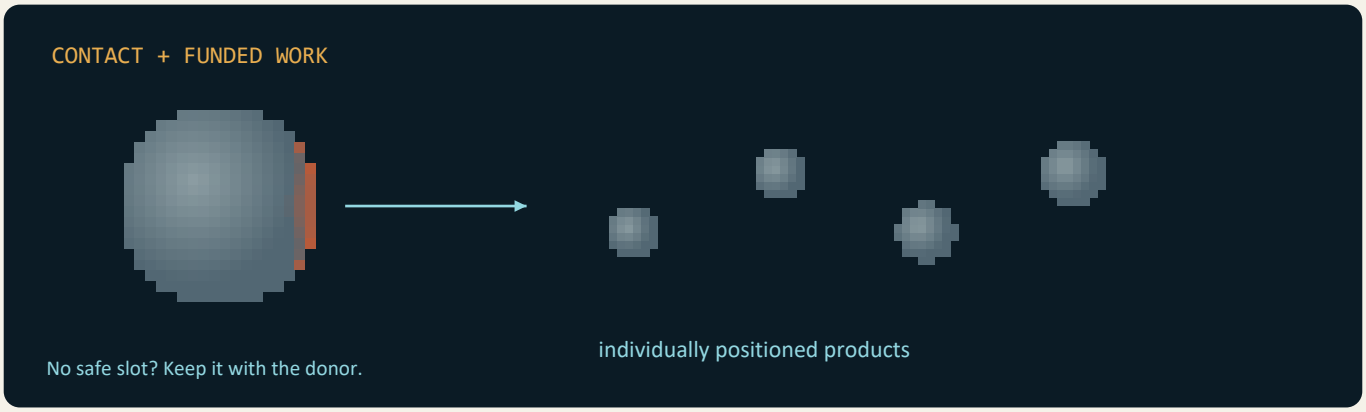
Important exclusions

bookGravityChange and envelope shedding retain their own loops because their skip rules differ. A shared helper is not permission to silently unify different transactions.

Review the units before the algebra
A compact release equation sometimes writes Φ for a pair term. Distinguish energy-valued helper output from specific potential before comparing derivatives or totals. This notation clarification changes no engine law.

Contact is a physical event, not a glow

Swept exteriors decide whether the existing outcome handler runs.



Products shown are individually positioned solids, not four decorative samples of an unseen rock population.

$$\mu = mM / (m + M),\ u = v - V,\ K_{rel} = \tfrac{1}{2}\,\mu\,|u|^2 \qquad 10.1$$

- **A miss:** no collision impulse. Crossing a gas envelope is not striking solid ground.
- **A supported merger:** masses add; actual momentum, spin, retained heat and material inventory pass through the existing transaction. Size follows retained volume.
- **Integration correction:** the owed mutual-gravity half-kick is applied reciprocally before contact and removed if the event is refused. It is not extra impact heat.
- **Capacity:** validate funding, minimum size, non-overlap and a destination slot before committing a product.

No universal temperature or debris share
Relative encounter speed and supported material/class outcomes matter. Two equal bodies can merge and retain their material; every encounter is not required to fragment or eject an escaping moon.

One funded impact, many real pieces

19.9.1 supersedes the old blanket escape ceiling for eligible impact products.

$$N(>d) \propto d^{-2.85} \tag{11.1}$$

$$v(d) = v_{ref}(d/d_{max})^{-1/3}\, s,\ 0.8 \le s \le 1.25 \tag{11.2}$$

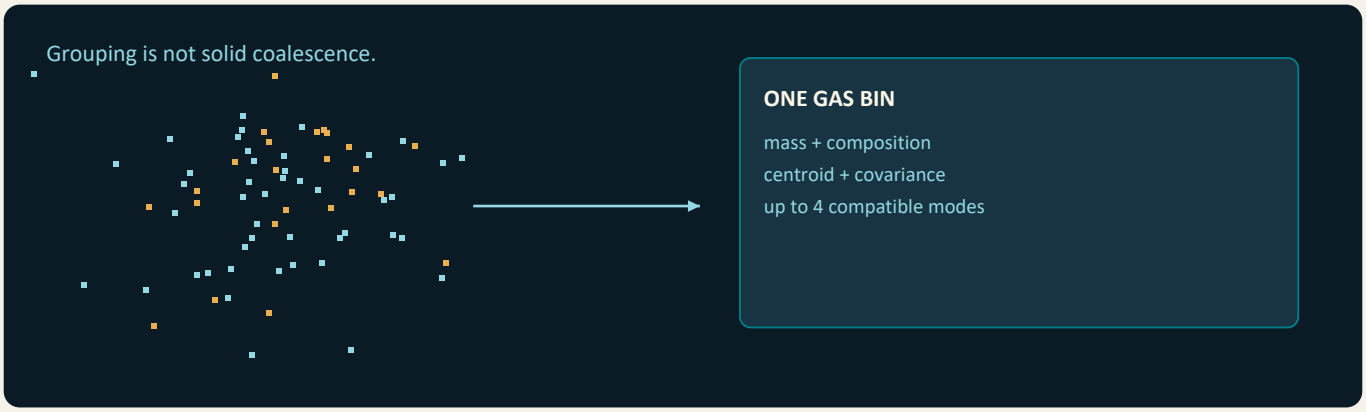
The recorded tail is a game-model family, not a claim of calibrated iron-impact predictions. Factor-two diameter bins hold real mass. Speed normalization uses the affordable tail, not merely the pieces admitted by the entity budget.

Question	Current interpretation
Where does each piece come from?	Debit actual donor mass and retained volume once. A cosmetic sample is not another physical owner.
Can it fit and be funded?	Use shared body slots, non-overlapping positions and the existing work ledger. Unspent material and work remain with the donor.
Can it escape?	Eligible curtain and recoil chips may escape when their actual launch is paid. Gravity decides fallback, orbit or escape; no extra kick or guarantee.
What remains restricted?	Belt-insertion windows and other retained legacy rules remain. This does not remove every launch restriction.

Birth floor versus safety margin
The physical minimum remains 1 km. The curtain sampler draws from 2 km to allow for measured contraction; undrawn mass stays with its donor. Existing hot pieces can still cross the cleanup cutoff later. See pages 38-39.

Packets and bins retain real state

A distribution record is neither an extra planet nor a physical solid merger.



$$P = \Sigma\, m_i v_i,\ c = (\Sigma\, m_i x_i)/M \qquad 12.1$$

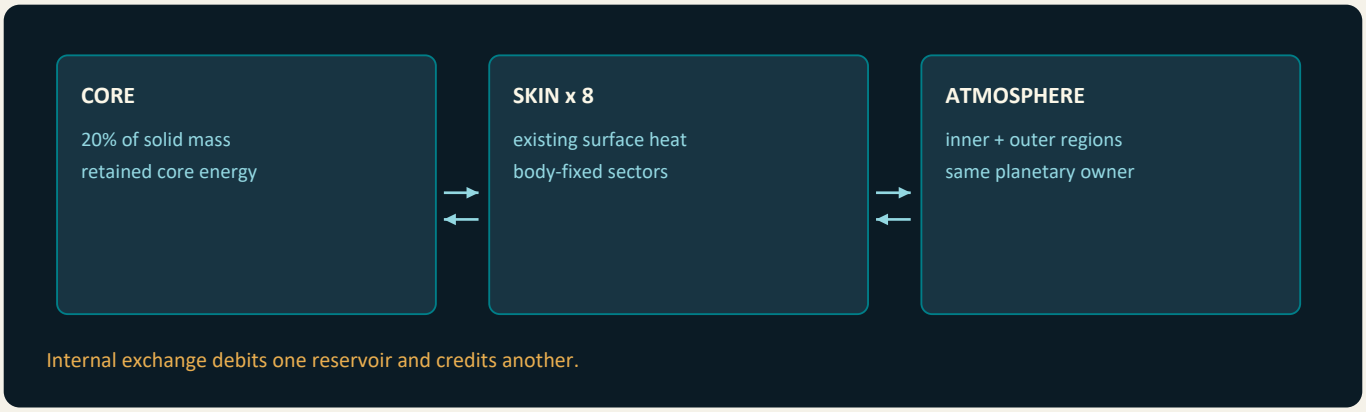
Bins retain mass, centroid, bulk velocity and second moments, plus up to four supported populations. Velocity spread carries kinetic energy; changing storage must not quietly turn that energy into heat or remove it.

- Population compatibility uses existing composition, dominant-phase and specific-heat criteria: approximately 5% composition and 2% heat-capacity matching, plus locality.
- Unsupported conservative transfers are refused and counted rather than forced into a different species or unrelated owner.
- Legacy distribution landings/encounters remain distinct from intact-solid contacts. An individual rock lands as an individual owner, not a fractional average.
- Loose deposits slide only above the supported repose condition. Released relief potential is booked as heat through the binding ledger, not a free cooling/heating multiplier.

Never hide solids in population modes
Earlier cloud-cloud solid stick/bounce/fragment descriptions are historical where the unified-solid architecture superseded them. Current supported gas/material modes are not an exception to the 1-km independent-solid rule.

Three locations, one retained energy total

The planetary skin is a coarse reservoir, not a measured microscopic layer.



$$U_s = \max(0, U_{total} - U_c - U_{atm}) \quad 13.1$$

$$U_{s,i} = w_i U_s, \Sigma w_i = 1 \quad 13.2$$

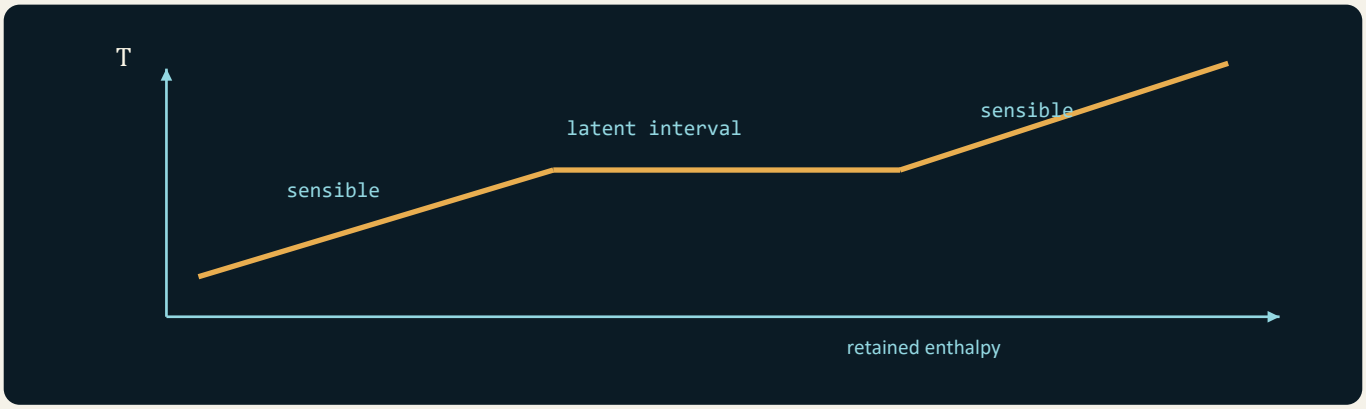
$$Q = (T_a - T_b)\,[C_a C_b/(C_a + C_b)]\,(1 - e^{-\Delta t/\tau}) \quad 13.3$$

Core mass share is 0.20 of condensed solid mass; the remaining skin mass is divided into eight regions. Pair relaxation debits one reservoir and credits the other. The formula above is the existing relaxation-time convention, not a newly calibrated surface-transfer coefficient.

Keep the local effect local
Each skin sector couples to its corresponding outer gas region. Later inner/outer and lateral transport redistribute heat. Do not replace a local impact with an immediate whole-atmosphere mean temperature.

Temperature comes from enthalpy

Phase energy is not a second copy of chemical formation energy.



Schematic enthalpy path. Width and temperature are not calibrated to a particular constituent.

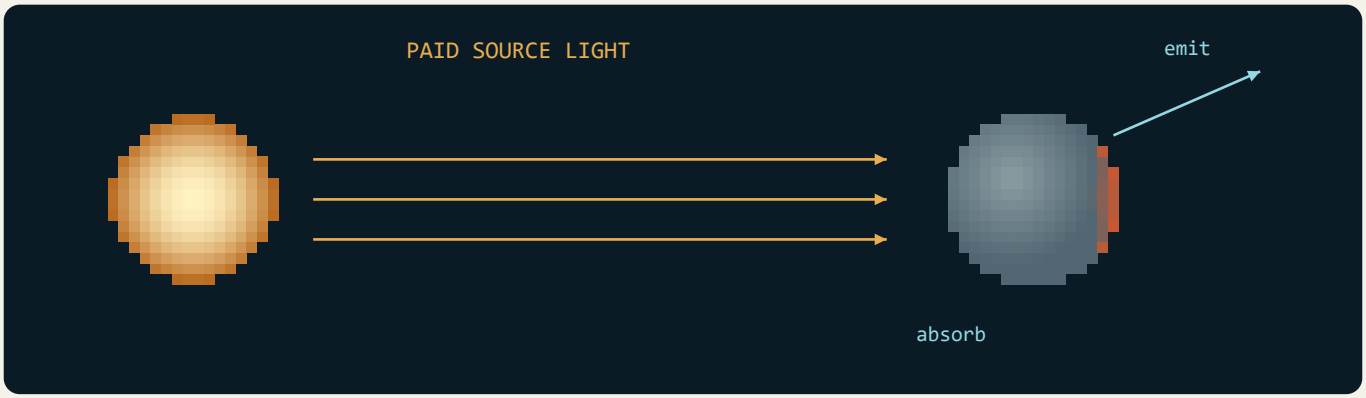
$$h(T,p) = \Sigma\, x_k \left[c_{p,k} T + L_k\, f_{melt,k}(T,p)\right] \qquad 14.1$$

$$r' = r\,(V'/V)^{1/3} \qquad 14.2$$

- Each supported constituent keeps its own pressure-dependent phase intervals. An Irio melting reference is not a universal material constant.
- Radius follows the changed condensed inventory volume. Required or released spin energy and pair-potential changes are settled through their existing ledgers.
- Water, molecular/compound and condensate inventories remain authoritative. A compound does not add mass above its constituents.
- A smaller condensed surface may mean material entered the same owner’s retained vapor. It does not automatically mean fragmentation or missing total mass.

Starlight supplies power, not a thermostat

The removed rapid relaxation must not return as an optimization.



$$P_{abs} = \alpha L A_{proj} / (4\pi D^2) \quad 15.1$$

$$P_{emit} = \varepsilon\sigma A (T^4 - T_{bg}{}^4) \quad 15.2$$

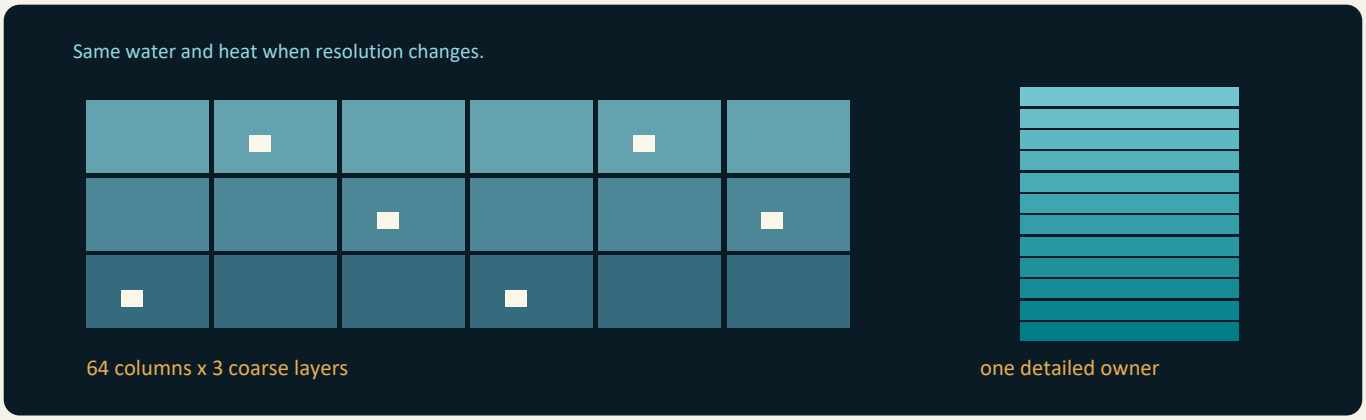
$$T_{rad} = [(\Sigma T_{s,i}{}^4)/8]^{1/4} \quad 15.3$$

A source pays its light through the existing ledgers and representable credit. Receivers absorb the paid share; the receiver’s own thermal model emits. A hot impact face can glow without gas. A hot core under cool skin does not by itself make the exterior incandescent.

Trace the clock branch
The existing 100x radiation clock, unresolved-wind boost of 10^5, and eligible gas 0.01 net-cooling fraction are separate conventions. Do not multiply every named constant into every object or accelerate gravity, age and chemistry.

Cloud appearance labels existing water

19.9.0 combines cloud water and vapour with the existing gas-layer raster.



$$H = R_{gas} T / (\mu_{molar} g) \qquad 16.1$$

- The weather column extends to at most five scale heights. Its hydrostatic layer weights, dry-adiabat temperature and radiative floor are coarse supported approximations.
- Formation lifts existing condensed water and pays that lift into binding. Settling/evaporation return lift; rain clears the cloud label.
- Live cloud work uses a 600-second universe interval and bounded owner updates. Coarse Stellar Time advances bulk state, not cloud speck history. Display combines cloud optical depth with the gas layer; it adds no material.
- Thermal planetary vents use gas-packet capacity even when the main-body table is full. Roche overflow keeps its body path. Gas giants use a cloud-deck reference, not invented solid terrain.

Resolution is not calibration
The model is water-cloud-only, uses a mean molar mass and bounded mixing, and has no full wind/climate solver. On a 12,000-km planet, 64 columns are roughly 590 km wide.

Drag conserves stored momentum

Small recoil must affect real future motion, not only a diagnostic total.

$$\ell = 1\text{-}e^{-\Delta t/t_s},\ \Delta p = -\mu \ell u \tag{17.1}$$

$$\Delta K = \tfrac{1}{2}\,\mu|u|^2\,\ell(2\text{-}\ell) \tag{17.2}$$

$$p = mv + r_p,\ K = |p|^2/(2m),\ \Delta x_r = r_p \Delta t/m \tag{17.3}$$

Encountered gas only

Exposure is clipped to actual passage and stops at exit or solid contact. Stopping laws depend on gas conditions and physical constituent size, not an atmospheric display ring.

Regime boundary

The Epstein branch uses mean free path greater than constituent diameter. Low-Re and inertial laws retain their existing joins. Four midpoint samples are an approximation with subdivision qualification.

Authoritative remainder

motion_remainder_kg_m_s stores SI momentum below the base velocity precision. Drift includes it, and normalization, split/merge and save/load carry it.

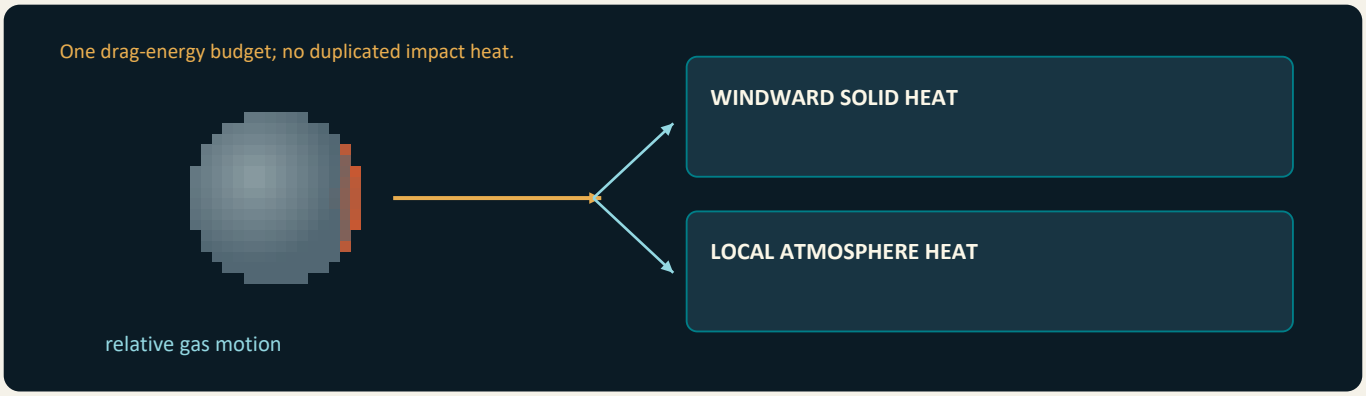
Energy consistency

The correction includes $v \cdot r_p + |r_p|^2/(2m)$. Bin velocity spread is not discarded as heat. The actual host receives reciprocal impulse.

Broad phase is not weaker physics
A conservative bound rejects proven distant misses, refreshes after state changes and keeps a reference path. It must not skip a real gas passage or contact to win a timing test.

Capped wall demand, then local gas heat

Entry is not a solid impact and does not spend the same energy twice.



$$q_c = 1.7415\times10^{-4}\sqrt{(\rho_g/r_n)}\,|u|^3,\ q_k = \tfrac{1}{2}\rho_g|u|^3 \qquad 18.1$$

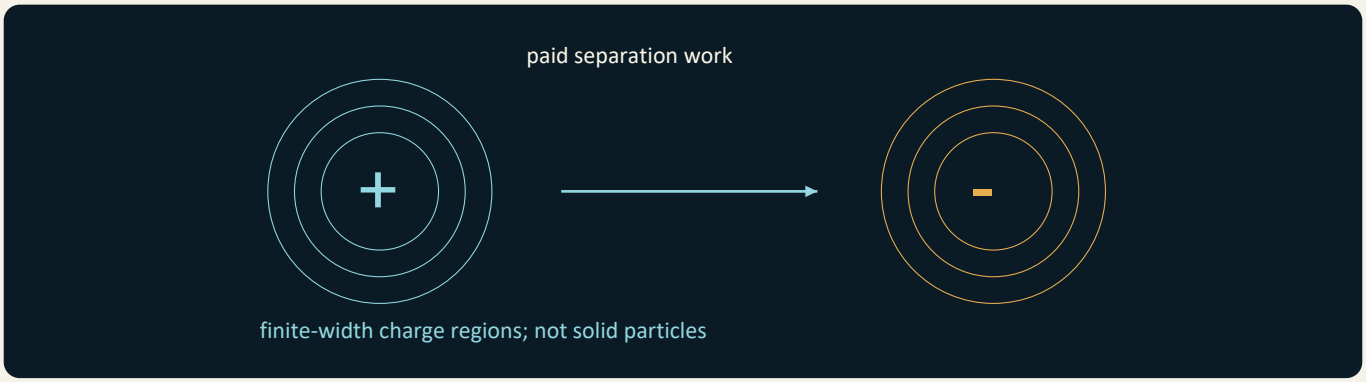
$$q = q_c q_k/(q_c+q_k),\ Q_s = \min(\Delta K, Q_{demand}),\ Q_g = \Delta K - Q_s \qquad 18.2$$

Demand includes constituent projected area, actual exposure, the supported carrier fraction, wall-enthalpy dependence and the smooth Mach join $(M^2-1)/24$. The kinetic-flux cap tends to zero in thin gas instead of abruptly switching all heat into the solid.

- New solid deposition follows relative gas motion into existing windward skin reservoirs; earlier heat remains attached to the body.
- Remaining dissipation enters encountered atmospheric sectors. Entry deposits no immediate core heat and creates no decorative physical gas parcel.
- This is a limited air-like convective gameplay proxy, not full shock radiation, calibrated Oxio chemistry or a universal planet-planet collision model.

Finite charge regions and paid work

The 19.8.0 review repaired cancellation for either sign of potential difference.



$$U = \tfrac{1}{2}\Sigma_{ij}\, q_i q_j K_{ij},\ K_{ij} = k\,\mathrm{erf}[r/\sqrt{(2(s_i^2+s_j^2))}]/r \qquad 19.1$$

$$\Delta U = A\,dq + B\,dq^2 \qquad 19.2$$

$$dq = 2W/[A+\sqrt{(A^2+4BW)}] \text{ for } A \geq 0 \qquad 19.3$$

$$dq = [\sqrt{(A^2+4BW)}-A]/(2B) \text{ for } A < 0 \qquad 19.4$$

A is V_{up}-V_{low}; B is $K_{uu}/2+K_{ll}/2-K_{ul}$. Gaussian width gives a finite self limit $k/(\sqrt{\pi}s)$. Work is paid from supported settling, equal/opposite charge is moved, and refused separation leaves its work as heat.

Gas discharges; solids store charge

White-blue-violet is a display convention driven by accounted release.

GAS DISCHARGE



stored charge alone
does not draw a bolt

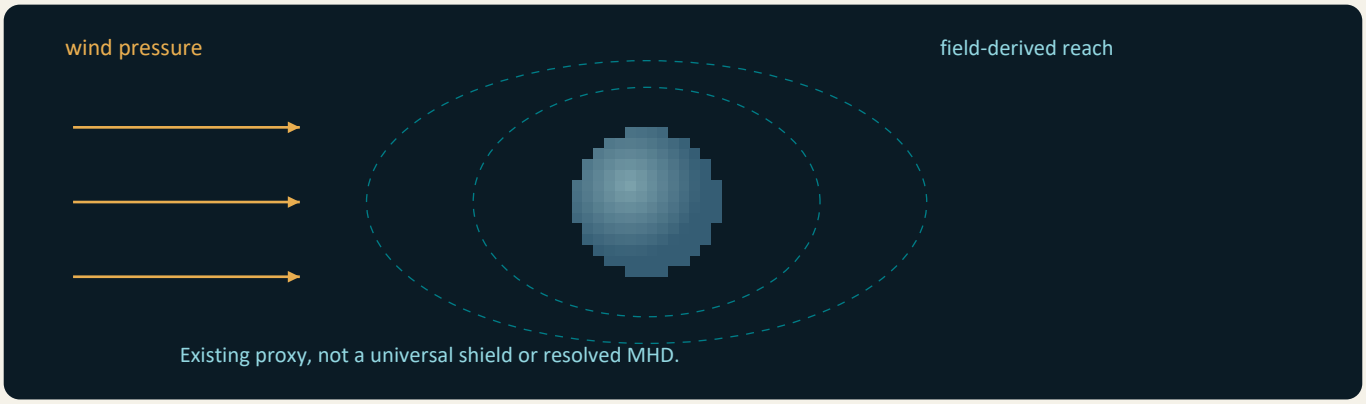
White core / blue glow / violet fringe. Schematic, not a spectrum.

Eligibility or parameter	Supported behavior
Charge generation	Mixed ice/supercooled-water settling only; at most 10^{-3} of released settling work. Heat/compression alone does not charge.
Conductors	Metallic Irio, Alumini and Gol; body-level percolation share 0.31. Charge does not automatically make a solid emit a bolt.
Breakdown row	Air-like N2/O2, about 120 Td; 180-1500 K, 0.01-10 bar, 1 m-1000 km gaps. No universal extrapolation to unsupported gas.
Threshold and release	Cloud enhancement 0.1; extinction 0.3 of threshold. Released energy: 1% light, remainder gas heat.

Not a voltage-to-colour spectroscopy solver
Display uses released energy, pressure and age, with a brief 0.35-second cosmetic fade. Stored charge alone draws nothing. Plasma/temperature alone does not guarantee lightning; solid incandescence stays separate.

Magnetism is not a universal shield

The equatorial field convention explains the pressure-balance prefactor.



$$B_{eq} = \mu_0 m_{dip}/(4\pi R^3) \qquad 21.1$$

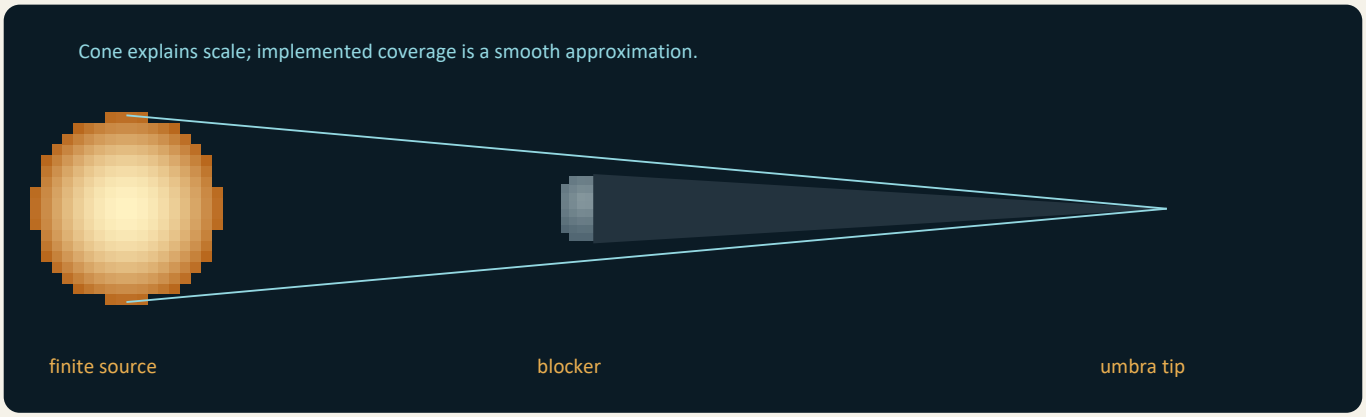
$$R_{mp} = R \max\{1, [2B_{eq}{}^2/(\mu_0 P_{wind})]^{1/6}\} \qquad 21.2$$

The recorded convention uses an equatorial field and boundary compression. It is not interchangeable with a generic $B^2/(2\mu_0 P)$ expression without checking the field definition. Raw balance and gameplay-limited reach remain separate.

- Moment is an inherited proxy from conducting interior eligibility, temperature/mass, spin and remanence; not a resolved MHD dynamo or a new field-energy solver.
- Coupling uses ionization/plasma proxies, not the net stored electrical charge. Magnetic outputs remain identical with and without separated charge in the compatibility fixture.
- Neutral debris still uses gravity, atmosphere and contact. Light follows radiation. No new terrain dipole force or immunity to impacts was introduced.

A finite source makes finite shadows

A tiny blocker does not become a planet-sized infinite shade.



$$t = [(b-s)\cdot(p-s)]/|p-s|^2,\ w = R(1-t) \qquad 22.1$$

$$\text{coverage} = \min(1, R_b^2/w^2)\ S(e),\ T = 1-\text{opacity}\cdot\text{coverage} \qquad 22.2$$

The shared helper uses blocker miss distance and a smooth edge weight $S(e)=e^2(3-2e)$, with bounds and interior guards. It is a finite beam-coverage approximation, not exact disk overlap or limb darkening.

- Display lighting selects at most eight casters. Shared code does not remove the need for consistent physical radii, opacity and interpolation at each call site.
- A receiving point inside the blocker is not a valid external-shadow test. A display glow halo is not new absorbing area.
- The familiar on-axis umbra length $D\ R_b/(R-R_b)$ is an explanatory scale when the source is larger. Finite receiver size and off-axis clearance still matter.

Compute a future state, not a replay

The requested date is a destination; it is not a list of ten-second frames.



What advances

Qualified orbital/free trajectories, bulk heat/material state, stellar age and supported consequential encounters advance toward the original requested endpoint.

What is omitted

No history of individual cloud pixels, lightning flashes, rendering interpolation, local shadow crossings or every tiny phase exchange is replayed. Appearance is reconstructed from final state.

What remains individual

Main-body mass, position, velocity and ID stay attached to their own owners. Group records organize calculations; they are not extra gravitational centres.

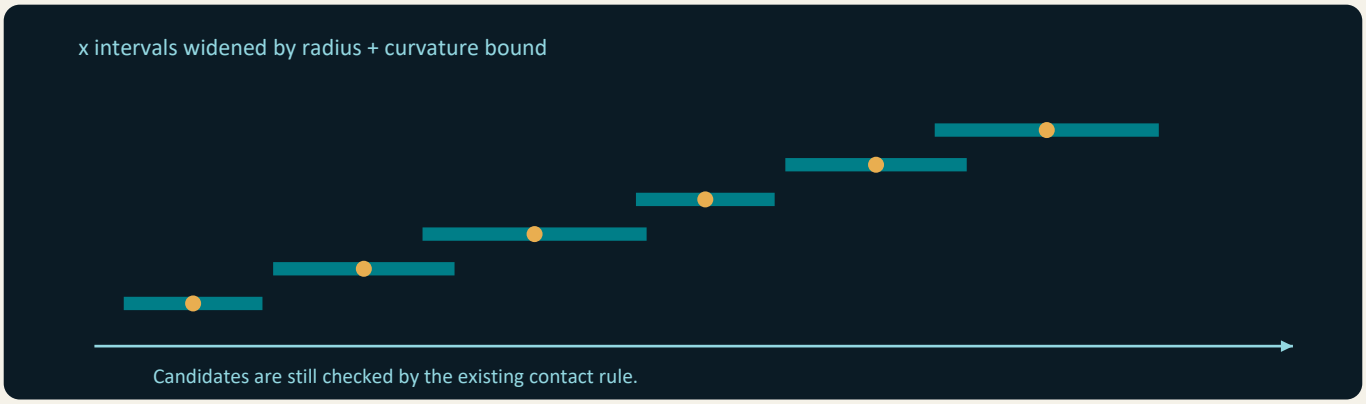
What a yield means

A work budget can return control and resume. It must not restart the request, duplicate accepted time or claim completion before reaching the requested endpoint.

Different numerical contract, same accounting obligations
Coarse mode is automatic for menu requests of at least 100 years; short qualifications can select it explicitly. Matching whole/subdivided tests must choose the same mode rather than accidentally comparing a different detailed controller.

REBOUND evaluates the game's forces

Faster candidate discovery is not permission to change a contact outcome.



- **Linked backend:** REBOUND 5.2.2, vendored commit c34134716cd484ed9cdb9f7d54ef8d859bdf547d. Trial positions invoke Kujira's force callback; stale owner references are rejected.
- **19.8.0 guard:** conservative x-sweep candidates replace the old all-pairs stage scan. Bounds widen by radius plus ½ $a_{max}\Delta t^2$; the all-pairs reference/fallback remains.
- **Outcomes:** predicted proximity is rechecked by existing physical handlers. Earliest predicted contact has priority, with distance/relative speed for ties; bodies do not merge merely because their orbital bands overlap.
- **Continuation:** important contacts and stellar boundaries update affected predictions, then the remaining skip continues. An initial internal step hint is not a permanent global ten-second cap.

> **Meaning of "exact" here**
> The force law is preserved in linked numerical work. Quiet analytic propagation and coarse encounter sampling remain declared approximations. Finite sampling does not prove exact reconstruction of every possible curved flyby or crowded microscopic contact history.

The populated request actually finishes

Fresh final-candidate completion logs, not a short or empty replacement scene.

Final 19.9.1 completion case	Wall time	Result
Full shared pools, 1 million years + 2.5 s	2.2797 s	14 windows; 0 retries; exact endpoint
Thermal population, 100 years	0.1125 s	17 bodies; heats/cools; exact endpoint
Original / memory-save continuation	1.7299 / 1.7146 s	Exact state/endpoint continuation
19.7 combined million-year scene	3.3997 s	15 windows; 1 merger; 7 events; 0 unresolved

The combined case

Full pools include an initially molten 64-km body, a closing 8-km / 16-km encounter and an escaping 325-km body. The supported handler resolves the encounter; retained material, identity, phase evolution, escape and the light ledger are checked. The 325-km test body is a legal size, not a new reference-scale constant.

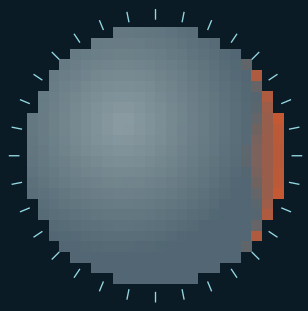
What linkage evidence proves

The final full-pool log records a real 60-second linked numerical window, 45 force evaluations and 1,600 tracked integration particles. Qualified quiet later windows are analytic. This is not a claim that every intervening orbit was numerically integrated for a million years.

> **Budget unchanged**
> The ReleaseSafe completion budget remains 30 seconds. Quietness must bound change over the remaining passive interval and invalidate after renewed light/impact/phase change. It does not force temperature to 10 K or freeze motion and aging.

History belongs to the surviving material

Deposits describe retained material; they do not mint another layer of mass.



BODY-FIXED RECORD

ID + birth tick
64 columns / 8 events
volume-neutral relief

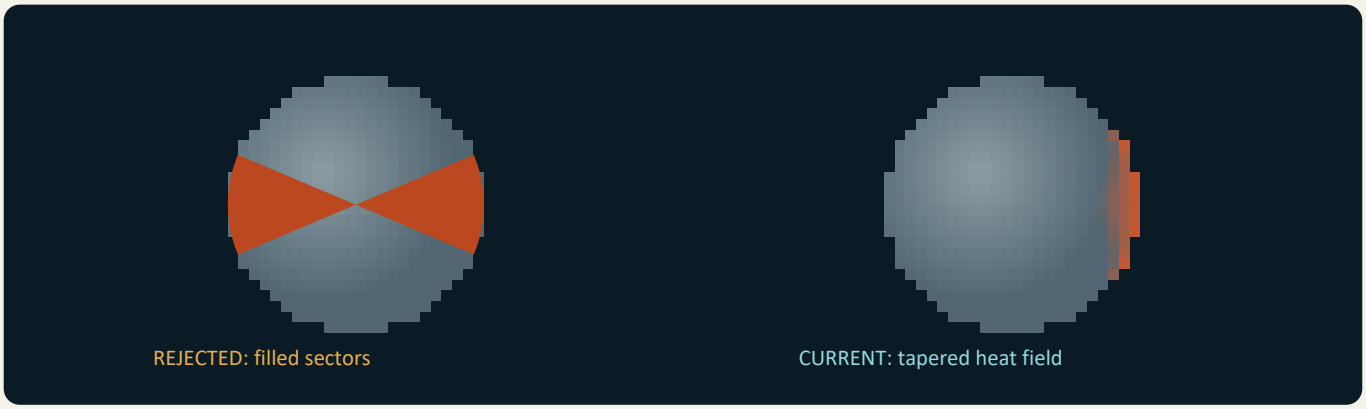
$$\Sigma w_i r_i^3 = \text{constant retained-volume measure} \qquad 26.1$$

- Records are keyed by ID and birth tick, with 64 columns and eight recent events. Mergers snapshot history and commit it after the physical solver.
- The footprint widens to fit the supported relief cap, using existing gravity/material strength and a 2% radius bound. Seeded relief uses 24 harmonics and a disposable 1,024-sample lookup.
- Seams and damage are descriptive outcomes modulated by melt and deposited specific energy. Heat colour is not allowed to deform the silhouette into a stuck-halfway merger.
- The water level is solved against retained liquid volume and the relief profile. A saved deposit depth is not a measured shock-penetration depth.

Current colour outranks old deposit RGB
Historical composition/seam colour can remain visible when cold, but it must not repaint a currently incandescent region as cool merely because the camera zoomed in.

A curved patch, not a filled triangle

19.6.5 reconstructs appearance from existing reservoirs. It does not simulate penetration.



Illustrative mask comparison, not gameplay screenshots or physical depth measurements.

$$e_i = \max(0, h_i - h_b),\ h_0 = \max(1\ \mathrm{J/kg}, h_b + L_f) \qquad 27.1$$

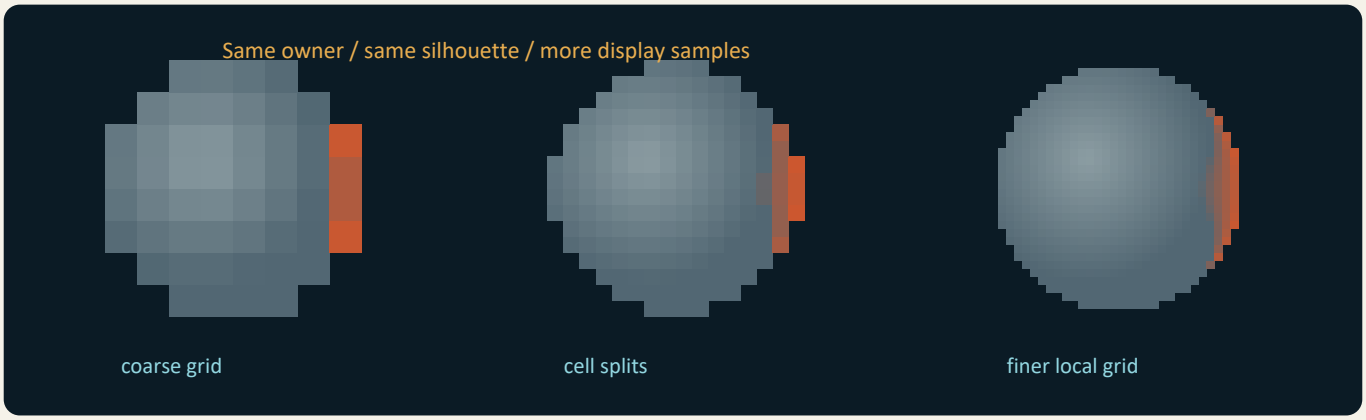
$$f_i = c\, e_i/(e_i + h_0),\ d_i/R = 1-(1-f_i)^{1/3} \qquad 27.2$$

Here c=0.8, h_b is the least-heated skin-specific enthalpy and L_f is the existing mixture latent fusion. Normalized positive cosine-to-the-fourth angular weights blend palette and depth. The depth mask fades below the actual drawn ground.

Depth is a presentation estimate
Surface-only depth is bounded by 1-cbrt(0.20), about 0.4152 R. Separately retained core heat can support central appearance; sector geometry alone cannot. Uniform hot solids stay opaque, and a cold core is not exposed through them. No new heat or atmospheric mass is created.

Smaller cells, the same owner

Refinement subdivides earlier cells rather than regenerating a new planet.



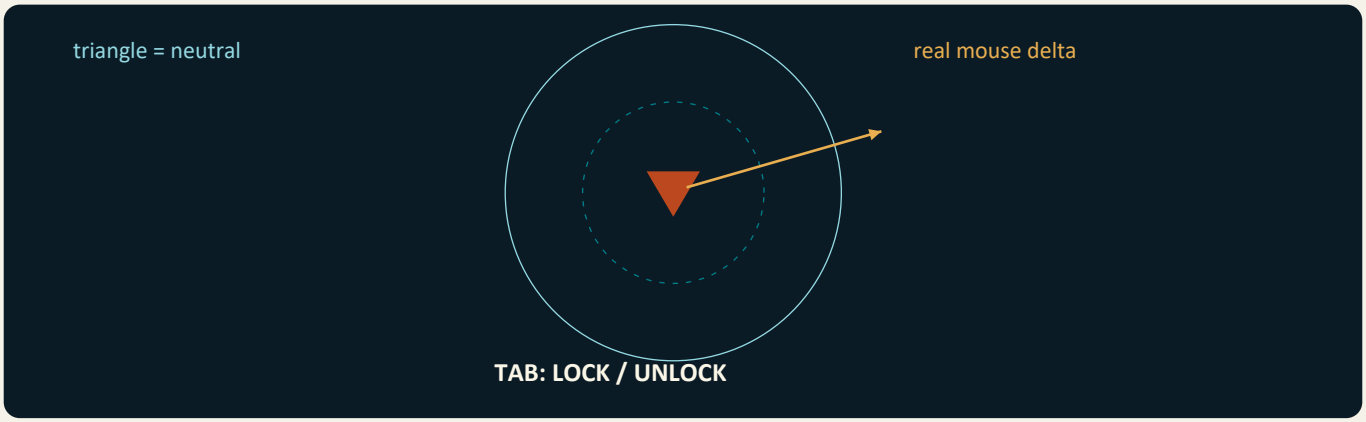
$$n = \mathrm{floor}[2 \log_2(\mathrm{zoom}/48)], \mathrm{cell} = 375\ \mathrm{km} / 2^n \qquad 28.1$$

The supported floor bounds n; the reported finest cell is about 1.46 km. This is quantized pixel refinement: each factor √2 in zoom halves cells and preserves earlier edges. It is not a promise of identical screen pixels at every wheel increment.

- Both normal and refined rasters use the same body-local thermal field and actual interpolated drawing angle. Compare the same material patch after rotation.
- The 10-km navigation triangle has a subpixel bracket; its exterior clearance covers its full footprint over drawn cells. Gas giants use their envelope floor.
- Unresolved averaging/fading and legitimate subpixel disappearance remain. Do not enlarge a physical rock or force a marker to satisfy an on-screen assertion.

Tab locks the same retained focus

The triangle is the steering neutral point; camera motion is not input.



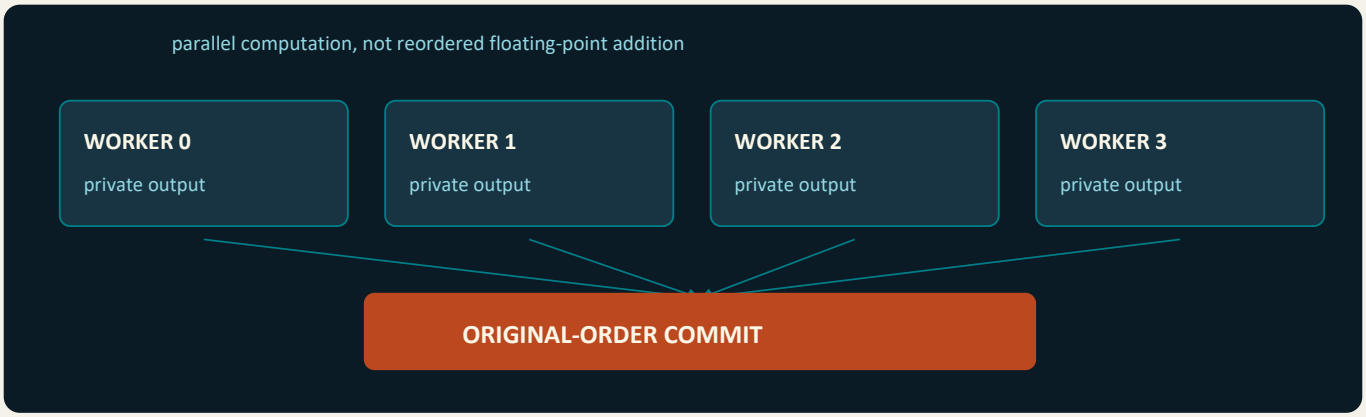
600-px reference setting	Value
Dead zone / full deflection	24 px / 260 px
Tangential / altitude speed caps	360 px/s / 180 px/s
Response lag / maximum altitude	0.12 s / 35% of viewport short side
Viewport scale	max(0.25, shortSide/600)

Selecting a body locks focus. Tab is the non-repeating keyboard twin of LOCK FOCUS / UNLOCK FOCUS. Pointer motion alone does not release the host; wheel zoom anchors on the triangle. Real input deltas feed a signed smoothstep and frame-rate-independent lag, projected onto the current surface tangent/normal.

Navigation is massless
Resize, menus and button interaction stop stale steering safely. The triangle is not a source of gravity, a collider, or a heat reservoir. The confusing old offset-neutral box must not become the input reference again.

Compute independently; commit in order

19.8.1 accelerated molten-heavy ticks without an approved physics approximation.



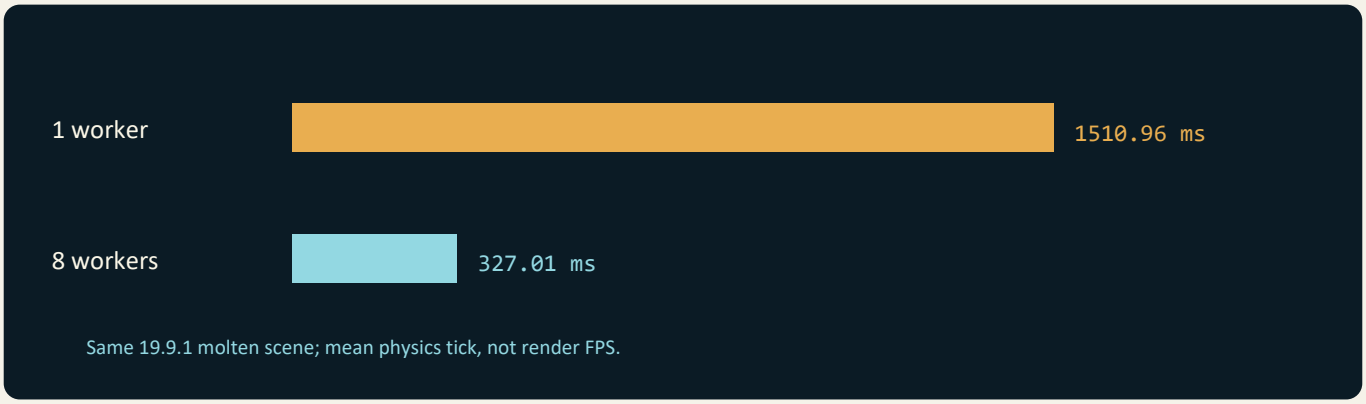
Parallel work	Sequential boundary
Per-source radiation rows; per-planet balances	Receiver sums, ledgers and commits keep original order
Long-time thermal owner estimates	Geometry/binding settlement and error reporting remain ordered
Pair-potential terms; packet/bin potentials	Body terms, then packet/bin terms summed in original order
Field-light source demand / receiver intake	Each internal sum keeps index order; source payment is sequential

The persistent pool uses up to eight workers within the CPU cap. The caller handles chunk 0. Helpers spin briefly, then sleep on the job counter; each acknowledges every job before the next is written. Nested or concurrent requests run inline. No fast-math or reassociation was added.

> **Agreement is measured, not a universal proof**
> Four molten ticks agreed bit for bit in the reported one/eight-worker comparison, including pre/post source versions. This qualifies the named fixture; it is not proof of every possible future thread schedule or world.

The molten case is faster, not free

19.9.1 final gate: affinity FFFF, the 16 performance-core threads of the i7-14700F.



Same 19.9.1 molten scene; mean physics tick, not render FPS.

Final 19.9.1 measurement	Recorded value
Ordinary full-pool mean tick, 1 / 8 workers	117.20 / 94.64 ms
Molten full-pool mean tick, 1 / 8 workers	1510.96 / 327.01 ms; about 4.62x
Full-pool render-only p95	154.9-166.0 ms; 250-ms stress limit
BH + 300 replay, worst play frame	81.6 ms; scene limit 250 ms
planet-gameplay-256, frame p95	371.3 ms; its separate limit is 375 ms

Physics-tick means, paused render p95 and active gameplay frames are different metrics. The full-pool render test records zero physics ticks; it cannot stand in for full-pool gameplay. Pinning and the quiet machine are part of the recorded conditions.

No universal smoothness claim
The BH replay improved from the recorded 19.9.0 worst frame of 91.1 ms, but this is one scene comparison. It does not guarantee 60 FPS or erase the slower crowded-gameplay result.

Save format 37, with explicit history

A reader keeps compatibility without preserving retired active solid physics.

Format	v19 addition
26-27	19.1 view contract; 19.2 sparse surface records and history
28-29	Material pools; loose layer and expanded 512-packet capacity
30-31	Pool thermal ledgers; sparse cloud ownership/state
32	Separated-charge regions and electrical ledgers
33-35	Sparse continuation and ownership-ID extensions
36	Authoritative momentum remainder; older supported layouts default it to zero
37	Shared-body solid geometry/lineage and within-capacity legacy conversion

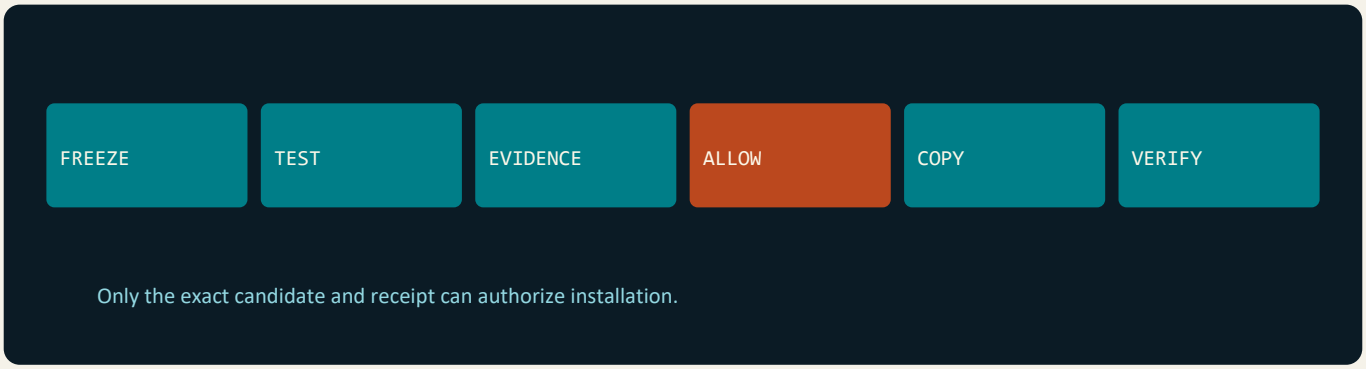
Current memory is not project disk usage

The reported Simulation is 38,356,992 bytes (about 36.58 MiB), an Entity is 3,640 bytes, and the full-pool save is 5,243,179 bytes (about 5.00 MiB). Compiler caches, logs, captures and rollback trees are separate disk artifacts, not live universes.

> **Atomic rejection is valid capacity enforcement**
> Within-capacity worlds must migrate and continue with valid IDs/references and required accounting. An oversized world is unsupported and must reject without changing input state. Newer saves do not become readable merely by rolling the executable back.

The final candidate earns installation

Installed 19.9.1; this documentation revision did not rerun the engine gate.



Final 19.9.1 gate	Recorded outcome
Focused contracts and launcher guard	All required exits 0; 19.9 physics 17/17
Full Debug	51/51 steps; 20 test programs cached; hard-solids program ran 33/33
ReleaseSafe rendering	1,241/1,243 passed; 2 skipped
Populated completion, fresh execution	9/9 with linked REBOUND
Debug / ReleaseSafe build commands	Passed using unchanged cached compile artifacts
Workspace / installed integrity	Expected 3 / 0; installed receipt verified

The frozen 206-file source matches the earlier run. That earlier full suite recorded 1,353 passing tests and a runner timeout in the hard-solids program, not a passing full run. The same program executed fresh in the accepted gate and passed all 33 checks. Original failures and labelled reused evidence remain visible.

Exact candidate and owner authorization
The guard binds source, binaries and acceptance evidence to the exact version/run authorization. The receipt was sealed, checked before copying and checked afterwards. A copied log is not a fresh execution; skipped tests are not passes.

What “v19 complete” does not mean

Completion is a release boundary, not a claim of a calibrated universe.

Declared approximations

Packet/bin mutual self-gravity is omitted. Thermal state has eight broad sectors and a 0.20 core share. Clouds are coarse, water-only and mean-molar-mass. Entry heating is air-like and lacks a full shock/ablation model.

Fields and optics

No inter-body Coulomb force or MHD solver is claimed. Charge regions do not drift with transport. Breakdown is limited to sourced air-like conditions. Shadows use a finite-coverage proxy and at most eight display casters.

Long-time model

Qualified quiet motion is analytic between numerical windows. Coarse encounter history is approximate; the symmetric central-tracer IAS15 conditioning case remains unqualified.

Visual model

Subsurface colour is a presentation of existing heat, not measured penetration. Relief is bounded and composite contact-binary bodies are unsupported. Sliding is rare at coarse column resolution.

Owner-approved cleanup

An isolated existing sub-1-km solid can be retired with its contents and energy. This explicitly nonconserving gameplay case is separate from ordinary conservative transactions.

Specific comparison dispositions

Exact recorded 19.6.4 heat/ID cases remain: phase heat about 0.44% versus 0.3%; core about 0.050% and skin about 0.012% versus 0.01%; one survivor-ID pair. These are not general tolerance increases.

Honest regression reporting
A passed gate does not turn the original approved-outlier comparisons into literal original-assertion passes. Independent conservation/reference checks still matter, and a new discrepancy is not automatically covered by an old exception.

Read the records behind the atlas

Local primary project records, not a new independent calibration study.

S01 / Complete v19 release

`docs/RELEASE_V19_COMPLETE.md`

Primary synthesis: milestones, models, capacities, persistence and limitations.

S02 / 19.8.0 M4 record

`docs/RELEASE_19_8_0.md`

Math review, charge root, contact sweep, exact parallel passes and combined case.

S03 / 19.8.1 exact-threading record

`docs/RELEASE_19_8_1.md`

Persistent pool, exact helper boundaries, agreement and before/after costs.

S04 / Fresh 19.9.1 completion evidence

`output/19.9.1/validation-final-20261029164106/completion-19.6.4.log`

Exact endpoints, linked work, nine tests and measured performance.

S05 / Final 19.9.1 gate and installed result

`output/19.9.1/validation-final-20261029164106/`

exits.txt, full.log, labelled reused log, rendering.log, gate-passed.json and post-install verification.

S06 / M3 model and release lineage

`docs/M3_HANDOFF_2026-10-27_19.6.5.md`

Detailed equations, source map, history, 19.6.5 masks and qualified exceptions.

H01: Universal Topology IV / 16.3.1, 52 pages.
H02: v16.6 Model Update Addendum, 18 pages.
H03: Planetary Surface and Subscale Physics Design, 12 pages.
All three are preserved under output/pdf/. H03 is a proposal, not installed behavior.

> **Source precedence**
> The final complete release and gate evidence supersede stale “candidate” headers and older stage blockers. External research names in source records identify model families; this PDF does not independently validate their calibration or every cited regime.

Keep the next review bounded

Use real reproductions, the current architecture and the same physical ownership.

Code family	Review connection
physics / planetary / material_phase	Contact transactions, local heat, phase settlement and conservation
material_packets / savegame	Gas/material modes, migration, authoritative recoil and IDs
deep_time / rebound_backend	Coarse scheduling, exact endpoint, motion guards and continuation
electric / magnetosphere	Paid field energy, stable root, supported breakdown and field convention
radiation / shadow_geometry	Source payment, receiving area, finite light and clock eligibility
surface / surface_view / raster / cloud_view	Retained history, bounded silhouette and shared subsurface colour
observation / parallel	Tab/triangle-centered input; persistent workers and original-order commits

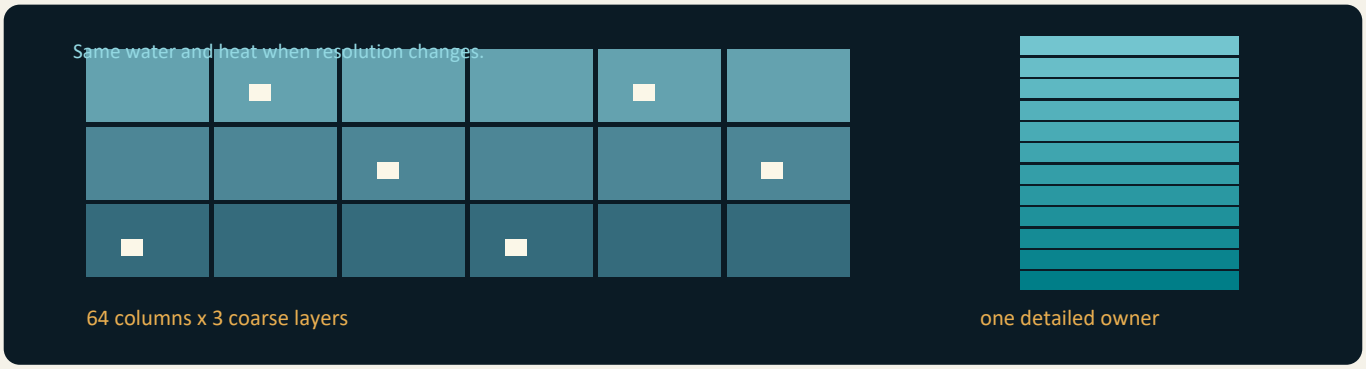
Useful next checks

- Compare matched force/potential derivatives, SI/world factors, relative motion and energy-valued pair ledgers.
- Test the same rotated surface patch and zoom state; do not change physical size or invent visibility to make a fixture pass.
- Compare actual stored momentum, phase inventories and worker-count agreement; a diagnostic balancing total is insufficient.
- Use the actual populated million-year and combined cases, not short replacements. Keep player saves and rollback evidence untouched.

Release boundary
Installed 19.9.1 closes the v19 release after M1-M4 and the 19.9 fixes. This atlas does not authorize a new physics subsystem, MHD/dust redesign, broader tolerance relaxation or an unguarded installation.

One atmosphere, not an opaque ring

Rendering reads retained gas and cloud inventories without spawning display material.



Schematic state: the installed display uses the gas-layer raster, not separate cloud blocks.

Correction	Installed behavior
Impact placement	Curtain and recoil products originate near the contact seam rather than wrapping evenly around the planet. Unplaceable material stays with the donor.
Thermal planetary vents	Ordinary vents enter the gas-packet pool, not a star-like main body. Packet capacity controls admission even when all 1,024 main-body slots are occupied.
Roche overflow	The existing gas-rich planetary stream toward its star keeps its body representation and supported encounter rules.
Cloud water and vapour	Cloud optical depth contributes to one gas-layer appearance. Thin vapour is drawn behind the limb and above the drawn ground, not as opaque blocks over the surface.

$$H = R_{gas}T/(\mu g),\ h_{draw} = \min(h_{envelope}, H\, f_{column}) \qquad 37.1$$

For vapour display, f_{column} = clamp[Σ/(1,000 kg/m^2), 0.1, 1]. A layer thinner than half a surface cell is not drawn at that zoom. Water vapour is pale blue at low opacity; ice is light blue and open water keeps its ocean colour.

Not a new physical envelope
The display uses existing temperature, column and scale height. It does not change physical atmospheric radius, pressure laws, material budgets or the stellar wind mechanism.

A population law, not equal chips

Actual solid volumes and the encounter ledger remain authoritative.

$$N(>d) \propto d^{-\beta},\ \beta = 2.85,\ a = 3-\beta = 0.15 \quad (38.1)$$

$$F_M(>d) = (d_{max}^{\ a}-d^a)/(d_{max}^{\ a}-d_{min}^{\ a}) \quad (38.2)$$

For common density, mass-weighting the recorded count tail gives this cumulative fraction. Factor-two diameter bins retain their own budgets. The few bodies admitted are mass-weighted samples; undrawn mass stays in its donor. This explains the documented model, not a new calibration.

Free shared body slots	Curtain pieces per donor event
512 or more	8
256-511	6
128-255	4
32-127	2
1-31 / zero	1 / zero

- Recoil planning retains its existing limit: at most four chips and the available slots. The planned mass is split into unequal pieces by the same tail, then normalized.
- The speed law uses the affordable full tail, not the subset drawn. Seeded scatter is 0.8-1.25; v_ref is at most half the actual collision speed and each piece is capped by that collision speed.
- The 0.20 launch share does not spend energy twice. Existing transactions must also fund lift, release, recoil, spin, gravity settlement and fragmentation before committing.

No rounding up or mandatory emission
The curtain draws from 2 km as a contraction margin; the physical minimum remains 1 km. Invalid, unfunded or unsafe births are refused. This tail and launch proxy are game approximations, not universal impact laws.

Paid escape is an outcome, not a switch

Recorded iron-body probes: a 0.29-solar-mass star, initially 60 K, at 20 million km.

Impact setup	Pieces	Diameters	Launch speeds	Planet escape at birth	Star escape
4,500 + 4,500 km; 5 km/s	12	3-387 km	2.1-8.5 km/s	4	0
6,000 + 6,000 km; 5 km/s	12	2-446 km	2.0-9.5 km/s	1	0
1,500 into 6,000 km; 15 km/s	12	2-403 km	0.8-9.1 km/s	4 chips	0
1,500 into 6,000 km; 30 km/s	11	2-456 km	1.7-19.4 km/s	3 chips	2

What the measurements establish

- Curtain pieces span the tail; resolved chips are no longer four equal 371-km bodies. At 15 km/s the four chips leave the planet but remain star-bound in the probe.
- At 30 km/s two chips also leave the stellar system in the reported continuation. Leaving a planet does not by itself imply leaving its star.
- Eligible below-belt-mass recoil chips no longer receive the old four-radius cap. Belt insertion windows and other retained legacy rules are unchanged.

Instantaneous escape diagnostics have limits
The source compares relative speed with the planet/star escape estimate at birth; the star diagnostic ignores the planet. Those estimates alone do not prove permanent escape in a full interacting scene. Later continuation is separate evidence.

The cleanup exception remains visible

In the 20,000-tick 15-km/s probe, two hot 2-km curtain pieces vaporized part of their solid inventory and crossed below the 1-km cutoff. Cleanup removed their owned contents: about 4.5e13 and 6.0e13 kg, roughly 1e-10 of the planet. The 2-km draw margin reduces contraction losses; it cannot prevent every later hot-body cleanup.

Not conservative vaporization
The physical phase change and gameplay deletion are separate operations. The latter is the previously approved narrow material/energy-loss exception, not an unreported relaxation of general conservation.

The exact release, with its provenance

19.9.1 installed 29 October 2026 at 17:35 CDT; this PDF performed no installation.

Release identity	Recorded result
Final gate	19.9.1 / validation-final-20261029164106
Receipt and install	Sealed; verified before copy and again afterwards
Installed integrity	0; executable, SDL2.dll and symbols match the receipt
Save format / capacities	37; existing body, packet and bin limits unchanged

Installed executable SHA-256

`C0514F98E72FA8126D4D40016F29AD48CEB4591A50BA13AC5456A2650C83CB69`

Rollback is local, not a public download

`checkpoints/19.9.1-before-20261029135225/installed-rollback/`

This retains 19.9.0. Rolling back an executable is not permission to open or rewrite player saves.

S07 / 19.9.0 corrective release

`docs/RELEASE_19_9_0.md`

Directional births, vapour packets and cloud/vapour display. The full-table vent admission regression is retained in the final gate.

S08 / 19.9.1 installed closeout

`docs/RELEASE_19_9_1.md`

Size/speed laws, paid chip escape, measured impacts, cleanup losses, cache provenance and installation identity.

> **Documentation-only publication**
> PDF and release/development notes only. Engine/game source, binaries, raw gate bundles, caches, player saves and rollback copies remain private. Copyright and licensing notices are unchanged.